Seokyung Park

M.S. Student | Pusan National University

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Personal Statement

M.S. student in Bio-Industrial Machinery Engineering with a background in Plant Bioscience. My research spans robotic gripping, quadruped-based environmental sensing, pile–soil interaction and structural monitoring. I aim to connect physical measurements with agricultural automation and plant phenotyping.

Qualifications

M.S. in Bio-Industrial Machinery Engineering
Pusan National University | March 2025 – February 2027 expected
GPA: 4.36/4.50

B.S. in Plant Bioscience
Pusan National University | March 2017 – February 2025
Cumulative GPA: 4.26/4.50 | Major GPA: 4.39/4.50 | Rank: 2/30
Summa Cum Laude; President's List

Appointments and Experience

Graduate Researcher
Precision Agriculture and Smart Sensing System Lab.
Pusan National University | March 2025 – present

Robotic gripping for fruit and vegetable harvesting

- Co-led study design and developed Python software for robotic compression experiments, force–displacement processing and hyperspectral analysis.
- Analysed tomato mechanics using sigmoid and viscoelastic models ($R^2 > 0.98$); compared four loading rates (6.9–15.3 N/s) and spectral changes at 1 h and 5 d after gripping.
- Responsible for software, validation, data curation and visualisation for a first-author Biosystems Engineering manuscript under review.

Uplift resistance of air dome foundations

- Modelled pile–soil interaction in Abaqus/Standard to evaluate wing length and inclination in calcareous sand; simulated 12.27–32.59 kN resistance at 7 mm uplift.
- Conducted a separate DIC pile pull-out study of localised soil displacement. Experimental validation of the X-Pile numerical model remains future work.

Structural sensing for air dome foundations

- Developed a load-cell acquisition setup using RS485, Raspberry Pi and CSV logging; conducted field measurements in Damyang in June 2026.
- Compared stationary acceleration-sensor noise; planned inclination sensing, MQTT communication and longer-duration structural monitoring.

Quadruped robots for agricultural sensing and field surveys

- Contributed to a Unitree Go2 platform with ROS 2 and environmental, LiDAR and inertial sensing for greenhouse mapping and rail-following experiments.
- Co-first-authored a greenhouse microclimate study across eight lanes, two heights and four sessions; the study produced 24 maps to assess fixed-sensor representativeness.
- Contributed to RTK-GNSS-linked reservoir surveys; repeated routes had a median nearest-path distance of 0.93 m, with 76% within 2 m.
- Investigated proprioceptive slope estimation across three surfaces and five slope settings: 1D-ResNet achieved 1.06° MAE and $R^2 = 0.92$ in five-fold validation on the controlled dataset.

Planned research: quadruped-based plant phenotyping linked to environmental measurements.

Undergraduate Research Student
Functional Crop Laboratory, Pusan National University | July 2023 – February 2024

- Used YOLOv8, Roboflow and ImageJ to analyse soybean leaf phenotypes under drought stress, with time-series RGB imaging and Arduino-based soil-moisture, temperature and humidity measurements.

Selected Development Project

Campus mobility, ride-sharing and dining app development

Designed and developed a cross-platform Flutter and Supabase app for Pusan National University Miryang Campus students, integrating train-bus transfer planning, bus timetables, real-time taxi ride-sharing, chat, fare settlement and campus dining information.
Implemented mobile architecture with Riverpod, GoRouter and Freezed; connected PostgreSQL, authentication, real-time updates, storage and Edge Functions; and built Python crawlers with Playwright and BeautifulSoup for campus information.

Publications

Published
Park, J., **Park, S.**, Jung, E., Song, J., Park, G., Lee, J., Kim, M., & Park, J. (2025). Inertial sensor applications in agriculture: Sensor types and data utilization techniques for mechanical and wearable applications. Korean Journal of Agricultural Science, 52(4), 447–469. https://doi.org/10.7744/kjoas.520405

Under Review
Park, S., Song, J., Hwang, H., Park, G., & Park, J. (2026). Speed-dependent control of robotic gripper for tomato harvesting to minimise compression-induced damage. Biosystems Engineering. Under review, second revision.

In Preparation
Hwang, H.†, **Park, S.**†, & Park, J. Greenhouse microclimate mapping using a quadruped robot to assess fixed sensor representativeness. Final preparation before submission. †Equal contribution.

Park, S., Park, J., Jang, J., Park, D., Kwon, S., Kim, M., & Choi, G. Analysis of uplift resistance characteristics of expandable piles in sandy soil with varying wing inclination and dimensions. Final preparation before submission; Korean manuscript, title translated into English.

Research Funding

Master's Student Research Encouragement Grant in Science and Engineering
September 2025 – August 2026
Development of Locomotion Algorithms and Autonomous Control Systems for Quadruped Robots Specialized for Korean Agricultural Terrain.

Conferences and Presentations

Titles originally in Korean are shown in English translation.

- Park, S. (2026, July 14). Height-resolved 3D microclimate mapping and spatiotemporal analysis within greenhouse crop canopies using a quadruped robot. AKABFE Student Research Spotlight, ASABE Annual International Meeting, Indianapolis, USA. Presenter.
- Park, S., Park, J., Hwang, H., & Kang, S. (2026). Ground-reaction-force-based lateral slope estimation for furrow-following quadruped robots. Korean Society for Agricultural Machinery, Spring Conference.
- Park, S., Park, J., Song, J., & Kim, M. (2025, October 31). Analysis of pile pull-out behaviour in Middle Eastern soil using digital image correlation. Korean Society of Agricultural Engineers, Autumn Conference, Jeju, Korea.
- Hwang, H., Park, S., et al. (2025, May 22). Remote control and integrated sensor-data environment for an agricultural quadruped robot. Korean Society for Precision Agriculture, Spring Conference. Coauthor.
- Park, S., Park, J., Park, G., Song, J., & Hwang, H. (2025, April 25). Minimising damage and optimising control in fruit and vegetable harvesting using a robotic arm and gripper with real-time force sensing. Korean Society for Agricultural Machinery, Spring Conference.

Honours and Awards

- Best Paper Presentation Award (team), Korean Society for Precision Agriculture, Spring Conference, May 2025; Autumn Conference, November 2024.
- Excellent Paper Presentation Award, Korean Society for Agricultural Machinery, Spring Conference, April 2025; Autumn Conference, October 2024.
- Excellence Award, PNU–PKNU Field Problem Research Group Final Symposium, November 2024; Best Presentation Award (team), Korean Society for Engineering Education, September 2024.
- Bronze Prize, College of Natural Resources and Life Science Capstone Design Competition, Pusan National University, October 2023.
- Excellence Award, Busan LINC 3.0 Entrepreneurship Shared University Competition, January 2023; PNU merit scholarships: four full-tuition and two partial awards, 2017–2024.

Technical Skills

Programming and analysis: Python, Dart and Flutter, NumPy, pandas, SciPy, Riverpod, GoRouter, Supabase and PostgreSQL; nonlinear model fitting, time-series processing and ImageJ.
Robotics and sensing: ROS 2, robotic manipulators and grippers, Unitree Go2, force sensing, environmental sensing, RTK-GNSS, LiDAR/IMU integration, Arduino, Raspberry Pi, RS485 and MQTT.
Imaging and modelling: RGB and hyperspectral imaging, SAM, YOLOv8, Roboflow, digital image correlation, Abaqus/CAE and Abaqus/Standard.

Professional Training

- Modeling Contact with AbaqusStandard, VPK PLM Solution Training Center, June 25–26, 2026.
- Introduction to Abaqus/CAE, VPK PLM Solution Training Center, March 18–20, 2026.